8 Features of manufacturing industries

8.3 Modern manufacturing methods and systems when designing for manufacture for small, medium and large-scale production

Topic	What students need to learn:
8 Features of manufacturing industries	8.3 – Characteristics, processes, application, advantages and disadvantages and the importance of considering accuracy of production and efficiency of modern manufacturing methods and systems when designing for manufacture for small, medium and large scale production: *a) production scheduling and production logistics* *b) robotics in production – robots on fully-automated production and assembly lines/cells*

Topic	What students need to learn:
8 Features of manufacturing industries	*c) materials handling systems – automated storage and retrieval systems (ASRS), automatic guided vehicles (AGVs)* *d) flexible manufacturing systems (FMS), modular/cell production systems* *e) lean manufacturing using just-in-time (JIT) systems* *f) standardised parts, bought-in components* *g) quick response manufacturing (QRM)* *h) data integration – product data management (PDM), enterprise resource planning (ERP) systems* *i) concurrent manufacturing.*

b) Robotics in Production

Robots on fully automated production and assembly lines/cells

Robotics in Production

- **Robotics** in manufacturing consists of computer-operated mechanical devices that can be programmed to carry out specific tasks.
- They usually take the form of an articulated arm fitted with a variety of ***'actuators'*** for carrying out operations such as welding, painting and assembly.
- They typically have ***six axes*** (degrees of freedom) of ***movement*** rather than the three (X, Y, Z) of most CNC machinery, in order to carry out complex processes.

Robotics in Production

- They play a key part in manufacturing cells to transfer parts from one stage to the next.
- The addition of a seven axis adds linear movement capability.
- Safety mechanisms and guards are an important feature of installations, but progress is being made in developing safer, autonomous robots, artificial intelligence (AI), proximity and force sensors and 3D cameras.

Advantages and Disadvantages of using Robots in Manufacturing

Advantages	Disadvantages
Powerful and efficient	Humans still required for servicing and fault diagnosis
No risk of repetitive strain injuries (RSI)	Expensive capital investment
No need for breaks and holidays	System/software problems can stop production
Easily programmed for a variety of tasks	High level of training required for programming
Adaptable due to different actuators	Potential issues with programming language conflicts
Capable of repetitive, precise operations	Loss of human skills and employment
Can carry out dangerous tasks such as working in toxic environments	Some limitations to tasks that can be undertaken
Ideal for welding processes such as MIG	Less suited to 'two-handed' assembly tasks Processes such as 'brazing' need human 'finesse' (skill/judgement)

Video: The Robot Revolution…



Now Test Yourself:

1. Explain the ***advantages*** and ***disadvantages*** of using ***<u>robots</u>*** to carry out an industrial manufacturing process of your choice.